Han Bao

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Personal Profile

Robotics researcher focusing on embodied navigation and robot learning. My prior research aimed to tackle coupled motion planning and decision-making via reinforcement learning, while exploring implicit environmental perception through representation learning. Currently, I am exploring how to adapt foundation models for safe, human-centric urban navigation. Additionally, my experience building a high-fidelity human-aware navigation benchmark has sparked a strong interest in RL-based character animation (DeepMimic, AMP) and humanoid control.

Education

2024 – present: MSc in Electrical and Electronic Engineering

Southern university of Science and Technology, Shenzhen
Supervised by Prof. Jiankun Wang.
Key Research Interests: Robot Learning, Embodied Navigation, Reinforcement Learning for Robotics.

2020 – 2024: B.Eng in Robotics

Zhejiang University of Technology, Hangzhou
GPA: 3.83/5.00 (Rank: 1/31).
Relevant Coursework: Introduction to Robotics:Mechanics and control, Modern Control Theory, Principle of Automatic Control, Signals and Systems, Machine Vision.
Thesis: Research on Robot Motion Planning Methods Based on Deep Reinforcement Learning in Dynamic and Complex Environments.

Publications

Learning Robot Visual Navigation in Crowds via Intention-Aware Scene Representations.

Han Bao, *Bingyi Xia, Hanjing Ye, Yu Zhan, Hao Cheng, Baozhi Jia, Wenjun Xu, Jiankun Wang*
IEEE Robotics and Automation Letters [RA-L 2026].

NavIsaacLab: Generating Realistic Crowd via Parallel Robot Learning for Benchmarking Human-aware Navigation.

Bingyi Xia, ***Han Bao***, *Jingyu Zhu, Hanjing Ye, Yuhan Pang, Guangcheng Chen, Liang Lin, Zhengyou Zhang, Wenjun Xu, Jiankun Wang*
Under review(Equal contribution)[PrePrint].

Exposing the Long-tail in Embodied Urban Navigation via Scalable Learning from In-the-Wild Videos.

Bingyi Xia, ***Han Bao***, *Zhewei Chen, Hanjing Ye, Jingwen Yu, Yuhan Pang, Wenjun Xu, Jiankun Wang*
Under review[PrePrint].

Projects/Research

Robot Urban Navigation Foundation Model | 2025.12-now

Shenzhen Key Laboratory of Robotics Perception and Intelligence (rπ Lab), SUSTech

- We extend prior research on human-aware robot navigation to more general scenarios, aiming to develop efficient and generalizable Physical AI that aligns with humans in complex urban environments.

Human-Aware Robot Navigation | 2024.03-2025.09

Shenzhen Key Laboratory of Robotics Perception and Intelligence (rπ Lab), SUSTech

- Developed a high-fidelity human-aware robot navigation benchmark platform based on IsaacLab. It generates crowd behaviors with realistic motions using character animation control methods, and provides multi-scale real-world scene assets along with baseline navigation algorithms.
- Designed a novel robot visual navigation method in crowds that utilizes BEV (Bird's Eye View) encoding and human pose embedding technologies to derive intention-aware scene representations. This representation significantly enhances the training efficiency of DRL-based robot navigation, achieving state-of-

the-art (SOTA) performance.
- **[Project Homepage 1] [Project Homepage 2]**

Vision-Guided Pan-Tilt Laser System | 2023
First Prize of Zhejiang Province, National Undergraduate Electronics Design Contest
- Developed a machine-vision-based pan-tilt control system that actuates a laser pointer for precise tracing of black tape boundaries and real-time dynamic tracking of a moving colored laser spot.

Vision-Guided Self-Balancing Bicycle | 2022
National First Prize, 17th National Undergraduate Smart Car Contest (Balancing Bicycle Category)
- Engineered an autonomous self-balancing bicycle system driven by a reaction wheel for active posture control. By integrating a real-time machine vision pipeline for precise trajectory tracking, the platform achieved robust dynamic balancing and high-speed racing.
- **[Project Demo]**

Honors and Awards
- Outstanding Undergraduate Graduate of Zhejiang Province — 2024
- Zhejiang Provincial Government Scholarship — 2022, 2023
- First Prize (Zhejiang Province), National Undergraduate Electronics Design Contest — 2023
- National First Prize, 17th National Undergraduate Smart Car Contest — 2022

Skills
- Programming: Python, PyTorch, C/C++
- Robotic: ROS/ROS2, NVIDIA Isaac-Sim/IsaacLab

Academic Service
Journal & Conference Reviewer

IEEE Robotics and Automation Letters (RA-L)
IEEE Transactions on Automation Science and Engineering (T-ASE)
IEEE International Conference on Robotics and Automation (ICRA)
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)